

LAB WORKBOOK

Fifty-two reusable briefs for planning, executing, validating, documenting, and safely tearing down hands-on work.

OUTCOMES

SLOs - safety - cost - user value

AI WORKLOAD

training - inference - retrieval - agents

PLATFORM

pipelines - serving - orchestration - registry

RUNTIME

frameworks - containers - drivers - libraries

FOUNDATION

compute - network - storage - facilities

52 WEEKS	365 NUMBERED TOPICS	ZERO TO HERO
52 lab briefs	Professional resource edition	Concepts + practice + evidence

EVIDENCE OVER ACTIVITY

A command running successfully is not the learning objective. Each lab should produce an artifact, a measurement, an explanation, and a safe cleanup record. Use these pages as printable worksheets or as a template for a repository README.

Minimum evidence bundle

- Environment: date, region or device, versions, image digest, accelerator, and relevant quotas.
- Reproduction: clean setup steps, configuration, commands, and seed or input reference.
- Validation: expected behavior plus quality, latency, throughput, resource, reliability, or cost evidence.
- Operations: health checks, logs, metrics, failure injection or negative test, rollback or recovery.
- Security: identities, permissions, secret handling, network exposure, data sensitivity, and cleanup.
- Reflection: what bottleneck or risk appeared, what changed, and what you would test next.

Spin Up Your First AI VM

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Introduction to AI Infrastructure.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Set Up Ubuntu for AI Development

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Linux Foundations for AI Engineers.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Inspect identity, paths, permissions, processes, open ports, disk, memory, and logs before changing state.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Compare Cost & Performance Across Clouds

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Cloud Infrastructure Basics.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Containerize a PyTorch Model

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Containerization Foundations.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy a Model on Minikube

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Kubernetes Fundamentals.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Build a Data Ingestion Pipeline

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Data Storage for AI.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Run a Model on GPU with CUDA

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from GPU Hardware Deep Dive.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Capture utilization, memory high-water mark, throughput, latency, power, and error state.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Train ResNet Across Multiple GPUs

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Distributed Training Basics.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Track Experiments with MLflow

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Workflow Automation & Experiment Tracking.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

CI/CD Pipeline for Model Deployment

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from CI/CD for AI Models.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy MLflow on Kubernetes

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Advanced Kubernetes for AI.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Optimize Cloud AI Workload Costs

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Resource & Cost Optimization.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Configure Load Balancer for AI API

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Networking for AI Systems.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Serve an Image Classifier with FastAPI

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Model Serving Basics.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy a Model on Triton Server

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Advanced Model Serving.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Monitor GPU Cluster with Prometheus

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Observability in AI Infrastructure.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Build a Drift Detection Pipeline

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Model & Data Drift.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Track feature shift, prediction shift, delayed labels, segment impact, data quality, and retraining decisions.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Secure a Model Endpoint with Authentication

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from AI Security & Compliance.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Build a HA AI Inference Cluster

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Reliability & High Availability.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Define SLI/SLO, RTO/RPO, dependency failure modes, retry budgets, health semantics, backup restore, and game days.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Serve a Model Across AWS + GCP

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Multi-Cloud AI Infrastructure.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy YOLOv5 on Jetson Nano

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Edge AI Infrastructure Basics.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Optimize a Model with TensorRT

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Optimizing AI for Edge Devices.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Build a Mobile AI App with TFLite

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Mobile AI Infrastructure.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Build a Streaming Pipeline with Kafka

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Data Pipelines for AI at Scale.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy a Simple RAG Pipeline

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Generative AI Infrastructure - Foundations.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Fine-Tune a Small LLM with PEFT

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Generative AI Infrastructure - Advanced.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy Real-Time Object Detection

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Infrastructure for Computer Vision at Scale.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy a BERT Model with FastAPI

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Infrastructure for NLP at Scale.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy CLIP for Image Search

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Infrastructure for Multimodal AI.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Train RL Agent with Ray RLlib

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Infrastructure for Reinforcement Learning.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Train Transformer Across Multiple Nodes

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Large-Scale Training - Basics.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Train with DeepSpeed ZeRO-3

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Large-Scale Training - Advanced.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Build a Pipeline in Kubeflow

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Enterprise MLOps - Foundations.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Automate Drift Retraining with Kubeflow

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Enterprise MLOps - Advanced.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Quantize a Vision Model

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Optimization Techniques - Foundations.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Train with Mixed Precision

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Optimization Techniques - Advanced.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Train a Federated Model with TFF

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Federated Learning Infrastructure.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Record trust assumptions, privacy budget or protocol, client skew, dropout behavior, aggregation integrity, and utility loss.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Apply Differential Privacy in Training

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Privacy-Preserving AI.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Defend Against Adversarial Attacks

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from AI Infrastructure Security - Advanced.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Configure Multi-Tenant Cluster

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Multi-Tenant AI Infrastructure.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy AI Infra on Low Budget

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from AI Infrastructure for Startups.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Design Enterprise AI Infra Plan

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from AI Infrastructure for Enterprises.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Build Real-Time Fraud Detection Pipeline

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Infrastructure for Real-Time AI.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy AI Agent for Robotics Simulation

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Infrastructure for Autonomous Systems.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Analyze Case Study Infra Architecture

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from AI Infrastructure - Case Studies.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Design Future-Proof AI Infra

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Future of AI Infrastructure.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Mini-Project Review Sprint

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Pre-Capstone Prep - Review.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Capstone Project Proposal

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Capstone - Problem Definition.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deploy First Working Infra Prototype

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Capstone - Implementation Phase I.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Deliver Scalable Infra Deployment

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Capstone - Implementation Phase II.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Capstone Infra Walkthrough

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Capstone - Finalization.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	

Graduation & Certification Showcase

Objective. Build the smallest repeatable artifact that demonstrates the core infrastructure decision from Capstone - Presentation & Graduation.

SCOPE	What is inside this lab boundary? What is explicitly out of scope?
TARGET	Write one measurable success condition and one safety or cost guardrail.
INPUTS	List code, data, model, image, configuration, credentials, and infrastructure dependencies.
PLAN	Write 4-7 ordered steps, including a clean start and a teardown or rollback step.

EVIDENCE CHECKLIST

- ☐ Artifact is versioned and can be rebuilt from a clean environment.
- ☐ Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.
- ☐ One expected failure or invalid input was tested and documented.
- ☐ Screenshot, log excerpt, metric table, or benchmark result is attached.
- ☐ Permissions, secrets, exposed ports, and data sensitivity were reviewed.
- ☐ Cost-bearing resources were stopped or removed; cleanup evidence is recorded.

RESULTS

Metric / signal	Baseline	Result	Interpretation

REFLECTION

What surprised you?	
Where was the bottleneck or risk?	
What would you change in production?	
What is the next experiment?	